Subject: Control Systems (23EE1603) **Date: 31-03-2026**

Assignment No. 2: Innovative Method- Tool Based Learning

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Que1. Create the following transfer function using tf(num,den) command. H(s) = (s + 5)/(s^2 + 8s + 16) and then convert from tf model (polynomial form) to zpk model (factored form) using MATLAB

Solution:- Code:

```
num = [1 5];          % Numerator coefficients (s + 5)
den = [1 8 16];       % Denominator coefficients (s^2 + 8s + 16)
sys = tf(num, den);   % Create transfer function [cite: 12, 39]
sys_zpk = zpk(sys);   % Convert to zpk model [cite: 12, 39]
disp('Transfer Function H(s):');
printsys(num, den);
disp('ZPK Model:');
disp(sys_zpk);
```

Output:

```
Transfer Function H(s):
 num/den =

      s + 5
   --------------
   s^2 + 8 s + 16


ZPK Model:
  zpk with properties:

          Z: {[-5]}
          P: {[2×1 double]}
          K: 1
```

```
 DisplayFormat: 'roots'
       Variable: 's'
       IODelay: 0
     InputDelay: 0
    OutputDelay: 0
      InputName: {''}
      InputUnit:{''}
     InputGroup: [1×1 struct]
     OutputName: {''}
     OutputUnit: {''}
    OutputGroup: [1×1 struct]
          Notes: [0×1 string]
       UserData: []
           Name:''
             Ts: 0
       TimeUnit: 'seconds'
   SamplingGrid: [1×1 struct]
```

Que2. Two systems with transfer functions are mentioned G1(s) =1/(s+2) and G2(s) = s/(s^2+2s+2) Find the transfer function of the system defined by the two cascaded subsystems using MATLAB.

Solution:

```
 % Q2: Cascaded (Series) systems [cite: 21]
  sys1 = tf(1, [1 2]);           % G1(s) = 1/(s+2)
sys2 = tf([1 0], [1 2 2]);       % G2(s) = s/(s^2+2s+2)
sys3 = series(sys1, sys2);       % Calculate cascaded TF [cite: 21, 46]
disp('Transfer Function of Cascaded System (G1 * G2):');
sys3
```

Output

```
Transfer Function of Cascaded System (G1 * G2):
sys3 =

        s
  ---------------------
  s^3 + 4 s^2 + 6 s + 4

Continuous-time transfer function.
```

Que.3 Two systems with transfer functions are given G1(s) =1/(s+2) and G2(s) = $s/(s^2+2s+2)$
Find the transfer function of the system defined by the two subsystems with parallel connection using MATLAB.

Solution:

```
% Q3: Parallel connection [cite: 22]
sys1 = tf(1, [1 2]);
sys2 = tf([1 0], [1 2 2]);
sys_para = parallel(sys1, sys2); % Calculate parallel TF [cite: 22, 54]
disp('Transfer Function of Parallel System (G1 + G2):');
sys_para
```

Output:

```
Transfer Function of Parallel System (G1 + G2):
sys_para =

    2 s^2 + 4 s + 2
  ---------------------
  s^3 + 4 s^2 + 6 s + 4

Continuous-time transfer function.
```

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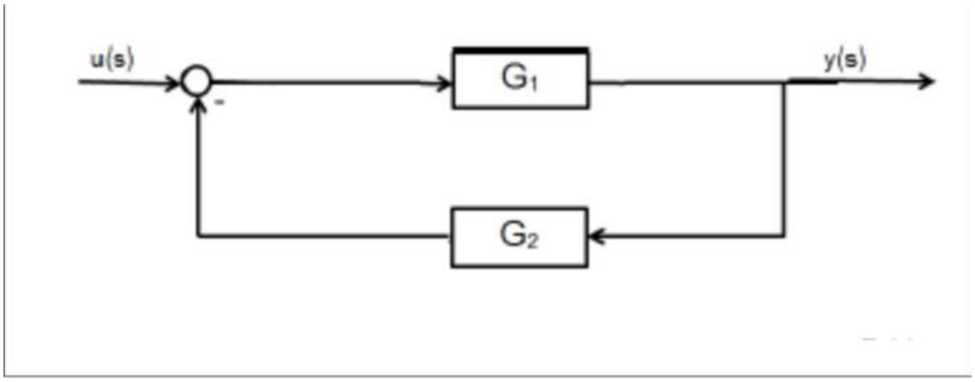
Que4. Given two systems with transfer functions G1(s) =1/(s+2) and G2(s) = s/(s^2+2s+2) Find the transfer function of the system with feedback elimination rule using MATLAB.



Solution:

```
% Q4: Feedback elimination [cite: 23]
sys1 = tf(1, [1 2]);            % G1
sys2 = tf([1 0], [1 2 2]);      % G2 (Feedback path)
sys_fb = feedback(sys1, sys2);  % Feedback elimination rule [cite: 23, 56]
disp('Transfer Function with Feedback:');
sys_fb
```

Output:

```
Transfer Function with Feedback:
sys_fb =

    s^2 + 2 s + 2
  ---------------------
  s^3 + 4 s^2 + 7 s + 4

Continuous-time transfer function.
```

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Que5. **For a given open loop transfer function of a system, plot Root locus using rlocus( ) command in MATLAB. Put K=48.**

$$G(s)H(s)=K/(s^3+6s^2+8s)$$

Solution:

```
% Q5: Root Locus Plot [cite: 28, 61]
K = 48;
num = [K];                  % Numerator is K [cite: 62, 63]
den = [1 6 8 0];            % Denominator: s(s^2 + 6s + 8) -> s^3 + 6s^2 + 8s [cite: 63]
G = tf(num, den);
figure(1);
rlocus(G);                  % Generate Root Locus plot [cite: 28, 61]
title('Root Locus for G(s)H(s) = 48 / (s^3 + 6s^2 + 8s)');
grid on;
```

Output: